

VEO Patent Strengthening Instructions

Non-Provisional Upgrade — Patent #3 — v2

Method and System for Establishing Portable, Verifiable Trust
Over Non-Deterministic Computation

Date: June 25, 2026

Scope: Expand protection from a specific implementation into a foundational infrastructure layer.

Rule: Do NOT rewrite the existing invention. Preserve all existing content and ADD the following concepts, embodiments, and claim coverage.

★ The Core Reframe: Trust Over Uncertainty

This is the most important instruction in this document.

The invention is NOT entropy.

The invention is **trust over uncertainty**.

Those are different things.

Current Framing Throughout the Provisional

- Generate entropy
- Score entropy
- Verify entropy

Required Framing Throughout the Non-Provisional

- Establish trust in non-deterministic computation

The Analogy

Standard	Did NOT Invent	DID Invent
PDF	Documents	Portable trusted documents
JWT	Identity	Portable trusted identity
VEO	Randomness	Portable trusted uncertainty

Concrete Instructions

1. In the **Abstract**, replace language about "generating entropy" with language about "establishing verifiable trust over non-deterministic processes."
2. In every **independent claim**, ensure the language describes establishing trust rather than merely generating or scoring entropy. The entropy is the input. The trust assertion is the output.
3. Add a section titled "**Trust Establishment**" or "**Trust Over Uncertainty**" that explicitly describes the system's purpose as creating independently verifiable trust assertions about non-deterministic computations.
4. Frame **ECS** not as "entropy quality scoring" but as "trust quantification for non-deterministic outputs" — this broadens applicability beyond randomness into any system where outputs involve uncertainty.
5. Frame the **verification framework** not as "checking an entropy object" but as "progressively evaluating the trust level of an uncertainty artifact."

The Test

A reader should finish the non-provisional believing the invention covers any system where:

- An output involves uncertainty or non-determinism
- A structured object captures the uncertainty's provenance and quality
- Cryptographic and/or immutable proofs establish trust
- Independent verification is possible without trusting the generator

That scope includes AI decisions, autonomous systems, simulations, governance, gaming — and anything else where non-deterministic computation needs trust infrastructure.

Goal: Protect the category, not just the implementation.

1. Reframe VEO as a Universal Uncertainty Artifact

Current framing: Verifiable Entropy Object is a structured entropy object.

Expand to: A Verifiable Entropy Object (VEO) is a **portable, machine-verifiable uncertainty artifact** that encapsulates one or more entropy sources, provenance information, quality assessments, cryptographic attestations, trust metadata, and verification evidence.

Goal: Protect the concept of uncertainty becoming a transferable, auditable digital artifact.

2. Add AI Decision Attestation Embodiments

The provisional mentions AI in the background but does not claim it.

Add embodiments covering: LLM tool selection, agent routing, multi-agent arbitration, resource allocation, autonomous task assignment, sampling decisions, autonomous execution approval, model selection, AI governance workflows.

Example claim language: A computer-implemented method comprising: generating a verifiable entropy object; associating the verifiable entropy object with an autonomous agent decision; recording a decision outcome linked to the verifiable entropy object; enabling independent verification of the randomness influencing the decision.

Goal: Directly cover future AI infrastructure.

3. Introduce Decision Receipt Concepts

New embodiment: Decision Receipt / Decision Attestation Record.

```
{
  "decision_id": "...",
  "decision_type": "tool_selection",
  "selected_option": "Tool B",
  "confidence": 0.81,
  "veo_id": "...",
  "signature": "...",
```

```
"anchor": "..."  
}
```

Covers: Human decisions, AI decisions, multi-agent decisions, governance votes, automated execution events.

Goal: Create continuation opportunities around auditable decision systems.

4. Generalize Signature Systems

Current: Ethereum, EIP-191, secp256k1.

Expand to: secp256k1, Ed25519, RSA, post-quantum signatures, future asymmetric systems.

Language: "Any asymmetric cryptographic signature scheme capable of proving authenticity and permitting verification."

Goal: Prevent easy design-around.

5. Generalize Hash Functions

Current: SHA-256.

Expand to: SHA-256, SHA-3, Keccak, Blake2, Blake3, future cryptographic hash functions.

Language: "One or more cryptographic hash functions."

Goal: Future-proof claim scope.

6. Generalize Anchoring Targets

Current: Blockchain anchoring.

Expand to: Public blockchains, consortium chains, private chains, append-only ledgers, transparency logs, notarization networks, distributed timestamping systems.

Claim: "Immutable verification substrate" instead of only "blockchain."

Goal: Protect the trust model rather than a specific ledger.

7. Verification as a Trust State Machine

Current levels: Structurally Valid, Cryptographically Verified, Anchored Verified, Policy Failed, Invalid.

Describe as: A trust state machine where an object transitions between verification states.

Additional states: pending, partially verified, expired, revoked, superseded.

Claim: "Assigning an object to one of a plurality of trust states based on progressively evaluated verification criteria."

Goal: Separate patentable value from the entropy itself.

8. ECS Beyond Fixed Weights

Add embodiments: Dynamic weighting, policy-based weighting, industry-specific weighting, machine-learned weighting, regulator-defined weighting.

Language: "The dimensions and weighting coefficients may be modified based on application requirements."

Goal: Avoid limiting protection to current percentages.

9. Interoperability Claims

Embodiments: One provider generates, another verifies, another consumes.

Key statement: "A verifier need not trust the original generator."

Goal: Standards-layer positioning.

10. Entropy-as-a-Service

Services: Entropy generation, VEO issuance, VEO verification, ECS scoring, anchoring.

Claim: "Issuing verifiable entropy objects through a network-accessible service."

Goal: Protect future API businesses.

11. Multi-Party Governance

Examples: DAO voting, lottery systems, regulated gaming, compliance approvals, corporate governance.

Pattern: VEO referenced during outcome generation; outcome remains independently auditable.

Goal: Increase licensing opportunities.

12. AI Audit Trail

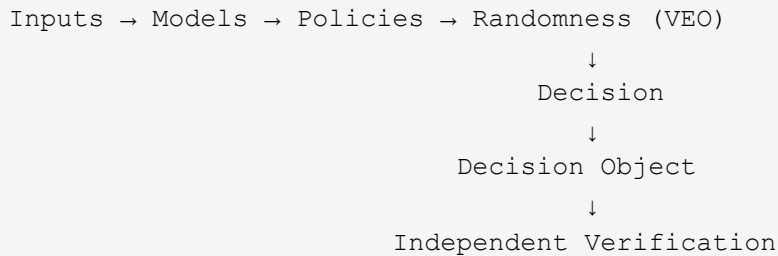
Covers: Decision provenance, replayability, regulatory review, model governance, post-event investigation.

Example: "A third party may reconstruct the decision context by verifying the associated VEO and related decision records."

Goal: Position VEO as future AI governance infrastructure.

14. Reserve Continuation: Patent #4 (VDO)

The non-provisional should explicitly leave room for a continuation covering **Verifiable Decision Objects (VDO)**.



VEO becomes one component of a larger decision-verification stack:

- VEO provides the entropy foundation
- VDO wraps the decision that used that entropy
- The VDO references the VEO, the model, the inputs, and the policy
- The entire chain is independently verifiable

Application domains for Patent #4: AI agents, autonomous vehicles, robotics, financial trading, healthcare workflows, enterprise automation.

To reserve this continuation: Ensure the non-provisional specification includes at least one embodiment describing a decision object that references a VEO, without fully specifying the VDO protocol. This creates the priority chain.

Strategic Summary

The non-provisional should no longer read as:

"A better random number system."

It should read as:

"A system for establishing portable, verifiable trust over non-deterministic computation — across software systems, AI agents, governance systems, gaming systems, and distributed networks."

Patent Family Progression

Patent	What It Protects	Category
#1	Generate verifiable randomness	Randomness generation
#2	Prevent randomness manipulation and preview	Randomness integrity
#3	Package into a portable, auditable trust artifact	Trust over uncertainty
#4 (future)	Use trust artifacts to verify AI and automated decisions	Decision verification

Each patent extends the previous one while opening a larger application domain. That progression maximizes long-term patent value, continuation opportunities, licensing potential, and acquisition attractiveness.

Reference Documents

#	Document	Purpose
1	Patent3_VEO_Provisional.pdf	Filed provisional application

2	RFC-0001-VEO1.md	Frozen protocol specification
3	ECS-v1.md	Frozen scoring specification
4	OpenRNG_Whitepaper_v1.md	Category-defining whitepaper
5	Agent Arbiter demo	Working AI decision attestation example